



VSCO RAW images

Photography app VSCO now supports RAW images for more phone photography potential. <http://dgit.in/VSCOraw>



Honor Magic

Chinese smartphone Honor's new Magic won't display notifications if it doesn't recognise owner. <http://dgit.in/HonrMgc>

Speaking of integration, it's only natural to question the efficacy of the lone sensor, working in isolation. So many kinds of useful insights can be gathered from aggregations of data of different kinds, and swarm sensors and Ubiquitous Sensor Networks (USNs) target an exploration of such potential.



By embedding spinach leaves with carbon nanotubes, MIT engineers have transformed spinach plants into sensors that can detect explosives and wirelessly relay that information to a handheld device similar to a smartphone.

Swarm sensors

Swarm sensors are now becoming the norm in large-scale surveillance. The idea behind swarms is to have a bunch of sensors working in communication and cooperation with each other so as to gather maximally useful information in the most efficient way. Simply put, instead of having robots work independently on the task at hand, which might lead to redundancies or wastage of resources, we train them to work constructively, eliminating waste. Swarms are today being used to survey agricultural land, forests, traffic routes, and even by space agencies in their strategies for extraterrestrial exploration.

Ubiquitous Sensor Networks (USNs)

USNs incorporate the integration of multiple kinds of sensors into one large information processing pipeline - this could include camera data, motions sensors, vibration data, sound and so on. USNs have tremendous applications in security and surveillance - with efficient integration, they approach the efficiency of a human guard, while limiting the cost, and widening the

area under surveillance. Like many technologies developed for defence, this has also leaked out to the entertainment industry - virtual reality is having a ball with the scope that USNs offer. VR now has the potential to react not just to your hand movement - but to your body temperature and facial expression that distinguish a frantic wave for help from a friendly one!

Self-healing Sensors

We come full circle with a return to the scope that biological inspiration offers sensing technology. Self-healing sensors are under rapid development. These are sensors that can actually sense and repair themselves post certain kinds of damage. The erstwhile comic book fantasy of an exoskeleton is thus under active development - this might well define the warfare of the 21st century. Self-repairing membranes for biological and chemical process plants, protective clothing for sports and adventures, as well as materials science for protective sheets, tarpaulins and roofs might eventually benefit from strides in this area.

Galvanic skin response sensors

Galvanic skin response sensors might appear to be straight out of science fiction. This remarkable technology senses various features of the skin, ranging from temperature to moisture, and has managed to effectively correlate these observations with emotion and mood. Besides the obvious applications to virtual reality (a mood or reaction-dependant plot twist in a movie or video game!), this creates the potential to monitor and study mental health and patient recovery in unprecedented detail. Similarly, it allows for the potential to tailor air-conditioning to individual preferences.!

EEG/EOG sensors

Electroencephalograms and electrooculograms (EEGs and EOGs) are a fascinating subset of sensing technology that track the electric fields produced by fluctuations in brain and eye activity

respectively. These have been linked to certain determinate processes that can be used to control all manner of processes. Tests have revealed that EEG data can be accurately used to learn and replicate hand gestures from human test subjects. Anywhere where human skill is required remotely - consider the example of underwater welding, which would be greatly simplified if a human could be sitting in a control room, directing a robot - will be revolutionised by this technology. EOGs offer a unique advantage - they track eye movements, and can be used in virtual reality, for attention measurement and psychological research and a host of other decisions. Consider data of where a driver's eyes flit while driving being used for the efficient design of vehicular control systems.

With the advent of multidisciplinary research, the possibilities for sensor research are truly endless. Add to this the interfacing, processing and learning capabilities that are being rapidly advanced, and you have a recipe for a



A remote sensing protective glove that can integrate galvanic skin response, self-healing, self-powered and other kinds of sensors. Photo for representative purposes.

future that will be as unrecognizable as it is unpredictable. Indeed, with the Internet of Things, that allows all your devices to interface with each other, with self-driving cars, with virtual reality - the world around us is already starting to look somewhat surreal. If we can expect a layman from the 1980s to be startled by the technology we take for granted today, it's only logical that humanity must be in for some big surprises in the 21st century.

For a technophile - there has never been a better time to be alive. 📱